Nidhi Parab

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Technical Skills

Languages: Python, Java, JavaScript, TypeScript, SQL
Frontend: React, Next.js, Angular, Tailwind CSS.
Backend & APIs: Node.js, FastAPI, Flask, Django, RESTful APIs, GraphQL, JWT, OAuth
Cloud & DevOps: AWS, GCP, Azure, Serverless (Lambda), Docker, Kubernetes, CI/CD (GitHub Actions)
Databases: MySQL, PostgreSQL, DynamoDB, NoSQL
Tools & Testing: Git, Linux, OpenShift, Jest, Cypress, Selenium

Professional Experience

Software Developer (Research Aide) — Sep. 2025 – Present
University at Buffalo — *Buffalo, NY*

- Developed AI4Ed and UB Grants Prospecting portals using Python, Django, Flask, React, and Docker, serving 100+ faculty and researchers.
- Built RESTful APIs with JWT-based authentication and Swagger docs, enabling secure, role-based access and improving onboarding efficiency by 40%.
- Optimized React frontends and Material-UI components, reducing load times by 35% and improving accessibility and usability.
- Performed automated scraping of grants from 5+ federal and external sources using Playwright and Selenium, feeding PySpark and cron-based pipelines for 25K+ records daily with 90% less manual effort.
- Automated summarization workflows using GenAI and LLM tools, enhancing data accessibility and researcher productivity.
- Implemented Airflow workflows for scheduled ETL pipelines, ensuring reliable, fault-tolerant data delivery across multiple sources.
- Containerized microservices with Kubernetes and OpenShift, achieving zero-downtime deployments and scalable, cost-efficient cloud operations.
- Authored technical documentation and API guides, enabling cross-team collaboration and faster integration.
- Collaborated in Agile sprints, conducted code reviews, and troubleshot production issues, improving operational reliability by 50%.
- Leveraged Tableau dashboards for data visualization, providing actionable insights and supporting grant decision-making.
- Reduced grant ingestion and summarization execution from 10 min to 30 sec via pipeline optimization and automation.

Data Engineering Intern — Jun. 2025 – Aug. 2025
Crypt0nest.io (Google for Startups) — *Austin, TX*

- Built backend and data infrastructure on Google Cloud Platform for a cryptocurrency ERP system within a startup environment.
- Designed 0→1 data architecture (Cloud Storage data lake → BigQuery Bronze/Silver/Gold warehouse) to support structured trading data and unstructured documents for RAG-based ML systems.
- Developed scalable ingestion pipelines in Python integrating CoinGecko and Coinbase APIs, processing 12M+ records with schema validation, rate-limit handling, and retry mechanisms.
- Implemented partitioned and clustered BigQuery tables with scheduled SQL transformations, producing production-grade financial and feature datasets consumed by the ML team.
- Automated containerized deployments using Docker and Cloud Build CI/CD, improving release reliability and reducing manual deployment errors.

Software Development Engineer — Oct. 2022 – Apr. 2024
Indian Development Foundation — *Mumbai, India*

- Developed and maintained a full-stack web application using Next.js and Node.js serving 300+ schools.
- Implemented secure authentication and authorization flows supporting multiple user roles.
- Built reusable frontend components and backend services following modular, maintainable design patterns.
- Worked closely with end-users to gather feedback, troubleshoot issues, and improve system reliability.

Projects

Personal Connections Memory App — Python, GraphQL, AWS Lambda, DynamoDB, S3 — Ongoing

- Developing a serverless application to manage professional networks, storing structured connection data and metadata.
- Implemented event-driven workflows using AWS Lambda and EventBridge to support scalable updates and notifications, ensuring reliability.

Google Colab Alarm Package — Python, JavaScript — Apr. 2025

- Developed and maintained an open-source PyPI library integrating frontend notifications with backend execution workflows.
- Applied clean API design and maintainable coding practices adopted by 1K+ users.

Education

University at Buffalo — *M.S. in Computer Science* — Aug. 2024 – Dec. 2025
Relevant Coursework: Algorithms, Machine Learning, Deep Learning, Operating Systems, Data Models, Computational Investment, Data Intensive Computing, Distributed Systems